Respuestas Trabajo Práctico 5

1. (a) Las pendientes de las rectas secantes son: $m_1 = 5$; $m_2 = 4{,}4$; $m_3 = 3{,}8$.
   (b) La pendiente de la recta tangente es 4.

2. (a) $f'(x_0) = 3$ (b) $f'(x_0) = -\dfrac{1}{x_0^2}$ (c) $f'(x_0) = \dfrac{1}{2\sqrt{x_0 - 1}}$

3. (a) $y = -2x - 3$ (b) $y = 3x - 2$ (c) $y = -\dfrac{2}{25}x + \dfrac{11}{25}$

5. (a) $f'(x) = 2x + 3, \quad x \in \mathbb{R}$
   (b) $f'(x) = -\dfrac{2}{\sqrt{x}} - \dfrac{2}{x^2}, \quad x > 0$
   (c) $f'(x) = -2x^{-3}\cos(x) - x^{-2}\operatorname{sen}(x), \quad x \neq 0$
   (d) $f'(x) = \dfrac{20x^3 \operatorname{sen}(x) - 5x^4\cos(x)}{\operatorname{sen}^2(x)}, \quad x \neq k\pi,\ k \in \mathbb{Z}$
   (e) $f'(x) = -\dfrac{4}{x^2}(\operatorname{tg}(x) + x^3) + \dfrac{4}{x}(\sec^2(x) + 3x^2), \quad x \neq 0 \text{ y } x \neq \dfrac{\pi}{2} + k\pi,\ k \in \mathbb{Z}$
   (f) $f'(x) = 3x^2\left(\dfrac{2}{\operatorname{sen}(x) + 1 + e^x}\right) + (x^3 - 5)\left(\dfrac{-2(\cos(x) + e^x)}{(\operatorname{sen}(x) + 1 + e^x)^2}\right), \quad x \in \mathbb{R}$
   (g) $f'(x) = \dfrac{(\cos(x) - x\operatorname{sen}(x))\left(\pi\sqrt{x} + \dfrac{\pi}{x}\right) - x\cos(x)\left(\dfrac{\pi}{2\sqrt{x}} - \dfrac{\pi}{x^2}\right)}{\left(\pi\sqrt{x} + \dfrac{\pi}{x}\right)^2}, \quad x > 0$
   (h) $f'(x) = e^x\operatorname{sen}(x)\sqrt{x} + e^x\cos(x)\sqrt{x} + \dfrac{e^x\operatorname{sen}(x)}{2\sqrt{x}}, \quad x > 0$
   (i) $f'(x) = \dfrac{2xe^x - (2 + x^2)e^x}{e^{2x}}, \quad x \in \mathbb{R}$

6. (a) Los puntos son: $(0, 20); (1, 15); (-2, -12)$
   (b) $y = \frac{\sqrt{2}}{2}\left(x + \frac{\pi}{4}\right) - \frac{\sqrt{2}}{2}$
   (c) $b = 0; c = 1$

7. (a) $\begin{cases} 2x, & \text{si } x > 0 \\ -3x^2, & \text{si } x < 0 \end{cases}$

   (b) $\begin{cases} \cos(x), & \text{si } x > 0 \\ 1, & \text{si } x < 0 \end{cases}$

   (c) $f'(x) = \operatorname{sen}(\frac{1}{x}) - \frac{1}{x}\cos(\frac{1}{x}), \quad \text{si } x \neq 0.$

8. (a) $f'(x) = -12(1+\cos^2(x))^5 \cos(x) \operatorname{sen}(x)$

   (b) $f'(x) = -\dfrac{1}{2}\left(\dfrac{1}{x+1}\right)^{\frac{3}{2}}$

   (c) $f'(x) = 7(3x^2+1)e^{x^3+x}$

   (d) $f'(x) = 10\sec^2\left((e^{2x}+1)^5\right)(e^{2x}+1)^4 e^{2x}$

   (e) $f'(x) = -\dfrac{1}{3}\cos(x^{-\frac{1}{3}})x^{-\frac{4}{3}}$

   (f) $f'(x) = \sec(x^2+\pi x)\operatorname{tg}(x^2+\pi x)(2x+\pi)$

9. (b) i. $f'(x) = \operatorname{sech}^2(x)$

   ii. $g'(x) = \cosh(x^2)2x$

   iii. $h'(x) = 2\cosh(x)\operatorname{senh}(x)$

10. (a) Recta Tangente: $y = 4\ln 2(x-2)+4$, Recta Normal: $y = -\dfrac{1}{4\ln 2}(x-2)+4$

    (b) Recta Tangente: $y = -\ln 3\ x + 1$, Recta Normal: $y = \dfrac{1}{\ln 3}x + 1$

11. (a) $y' = \dfrac{x}{y}, \quad y \neq 0$

    (b) $y' = \dfrac{y^3 - \dfrac{4y}{\sqrt{8xy}}}{2 - 2y - 3xy^2 + \dfrac{4x}{\sqrt{8xy}}}, \quad xy > 0, \text{ y } \ 2-2y-3xy^2+\dfrac{4x}{\sqrt{8xy}} \neq 0$

12. (a) $y = -\dfrac{3}{4}x + \dfrac{25}{4}$

    (b) $y = -x + 6$

13. (b) i. $y' = \dfrac{1}{3}x^{-\frac{2}{3}}$

    ii. $y' = \dfrac{1}{2\sqrt{x-1}}$

    iii. $y' = \dfrac{1}{x^2+1}$

14. (a) $y = \dfrac{1}{4}(x-3)+1$

    (b) $y = \dfrac{1}{3}(x-1)+2$

15. (a) $g'(x) = \dfrac{\left(\frac{5}{2}x^{\frac{3}{2}} - \dfrac{1}{2\sqrt{1-x}}\right)x \operatorname{arc\,sen} x - \left(x^{\frac{5}{2}} + \sqrt{1-x}\right)\left(\operatorname{arc\,sen} x + \frac{x}{\sqrt{1-x^2}}\right)}{(x \operatorname{arc\,sen} x)^2}$

(b) $f'(x) = \dfrac{\cos(2\sqrt[3]{x+1})\left(\frac{2}{3}(x+1)^{-\frac{2}{3}}\right)\cos x + \operatorname{sen}(2\sqrt[3]{x+1})\operatorname{sen} x}{\cos^2 x} - \dfrac{1}{2\sqrt{\arccos x}\sqrt{1-x^2}}$

(c) $f'(x) = 3^{4x} 4 \ln 3 \operatorname{arc\,tg}(\sqrt[3]{x}) + 3^{4x} \dfrac{\frac{1}{3}x^{-\frac{2}{3}}}{x^{\frac{2}{3}}+1}$

(d) $f'(x) = \dfrac{\frac{1}{4x}(1+\ln(x^2)) - \ln(\sqrt[4]{x})\frac{2}{x}}{(1+\ln(x^2))^2}$

16. (a) $f''(x) = 6x + 10$

(b) $f''(x) = \dfrac{(\frac{1}{4}e^{\frac{x}{2}} - \frac{1}{4}xe^{\frac{x}{2}})(1-x)^2 + (\frac{3}{2}e^{\frac{x}{2}} - \frac{1}{2}xe^{\frac{x}{2}})2(1-x)}{(1-x)^4}$

(c) $f''(x) = 6x\cos(x^3) - 9x^4 \operatorname{sen}(x^3)$

(d) $f''(x) = \dfrac{-18x^2 - 30}{(3x^2-5)^2}$

17. (a) Hasta el orden 2.

(b) $g'(x) = 6x^2 - 32x + 1, \quad g''(x) = 12x - 32, \quad g'''(x) = 12$

(c) $f^{(n)}(x) = (-1)^n\, n!\, x^{-(n+1)}, \quad f^{(n)}(1) = (-1)^n\, n!$
siendo $n!$ el factorial de n, definido como: $n! = n(n-1)(n-2)\cdots 2 \cdot 1$